Hein Zarni Aung

+1-(917)-854-7709 | heinzarniaung.dev@gmail.com | linkedin.com/in/heaung | github.com/LoistCloist

EDUCATION

Stony Brook University Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Software Security, Full Stack Development, Algorithms

INTERNSHIPS

Software Engineering Intern — Clustr Jan 2026 – Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows and activity feed endpoints.
- Implemented secure authentication features, including a password reset flow (client screen plus backend cloud function) and custom claims to drive new-user onboarding logic.

Software Engineering Intern — Prenora Dynamics LLC Jun 2026 – Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors, using the Supabase CLI to manage schema migrations and configure user auth/login flows.

RESEARCH

Undergraduate Research Assistant — Ethos Lab, Dept. of Computer Science, Stony Brook University Sep 2025 – Dec 2025

PI: Abisheka Pitumpe | Team: 2

- Built a pipeline to monitor newly issued SSL/TLS certificates in real time and flag potentially fraudulent domains impersonating EZ-Pass tolling services, scraping keywords from the raw HTML, CSS, JS, and URLs of newly created sites.
- Used Selenium to automate certificate and domain scraping and Python to filter and triage flagged sites, testing and monitoring hundreds of domains to separate real payment fraud from noise.

CAMPUS INVOLVEMENT

Teaching Assistant – Data Structures & Algorithms, Dept. of Computer Science, Stony Brook University Aug 2025 – Dec 2025

- Created original PowerPoint-based review materials from scratch for midterm exam preparation; materials were adopted as the basis for subsequent semesters' review sessions.
- Hosted 2-hour midterm review sessions reaching 100+ students combined in-person (~50) and via Zoom, communicating technical concepts clearly to a large, mixed-experience audience.

PROJECTS

TuneAcademy React, NumPy, Essentia, FastAPI, GitHub Actions

- Built the AI/audio-analysis pipeline for an app matching music students with instructors based on specific technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances.
- Set up a CI/CD pipeline using GitHub Actions to lint and test the codebase, then auto-build and deploy TuneAcademy to Cloudflare Workers.

SKILLS

Languages: Java, Python, JavaScript, C, TypeScript

Backend: Node.js, Express.js, FastAPI

Frontend: React, React Native, Next.js

Databases & Cloud: PostgreSQL, MongoDB, Firebase/Firestore, AWS, GCP, Supabase

Tools: Selenium, Git, Docker, Linux, PM2